

Narrative Consistency Verification in Long Documents

By: Team Unlabeled

TRACK A: Using LLM and Generative AI

INTRODUCTION TO PROBLEM STATEMENT

Input:

- A full novel (long narrative text)
- A hypothetical backstory for a character

Task:

- Determine whether the backstory is **consistent** or **contradictory** to the novel

Challenge:

- Evidence is **distributed across chapters**
- Contradictions are often **implicit**
- Reasoning requires **global narrative understanding**

Basic Approach 1: Giving Novels directly to LLMs

Expected it to be:

First intuition: use a powerful LLM (GPT)

Prompt includes:

- Entire novel
- Few-shot examples
- Backstory claim

Expectation:

- LLM should reason holistically

Ground Reality:

Full novel exceeds context window

Long narratives dilute causal signals

LLM reasoning becomes:

- Shallow
- Inconsistent
- Sensitive to prompt phrasing

Observed result:

- **~60% accuracy** on train.csv

Conclusion:

- Direct LLM prompting is **insufficient**

Basic Approach 2: Using Standard RAG pipelines

Thought it would do:

Applied classical RAG pipeline:

- Embed → Retrieve → One-shot LLM reasoning

Observations:

- Retrieval is **single-pass**
- Evidence is **discarded after one step**
- Long-range dependencies ignored

Ground Reality:

- Fails on global contradictions
- Overlooks temporal and causal structure

Scenarios where normal RAG pipeline and LLM struggles

Character Development Over Time

A backstory may match a character's early behavior but contradict who they become later in the story.

Example: A character portrayed as morally conflicted early is given a backstory describing lifelong cruelty.

Timeline and Causal Violations

A backstory may place events before they are possible or ignore cause–effect relationships in the narrative.

Example: A character is described as acting on knowledge gained only in later chapters.

Scattered Evidence Across the Novel

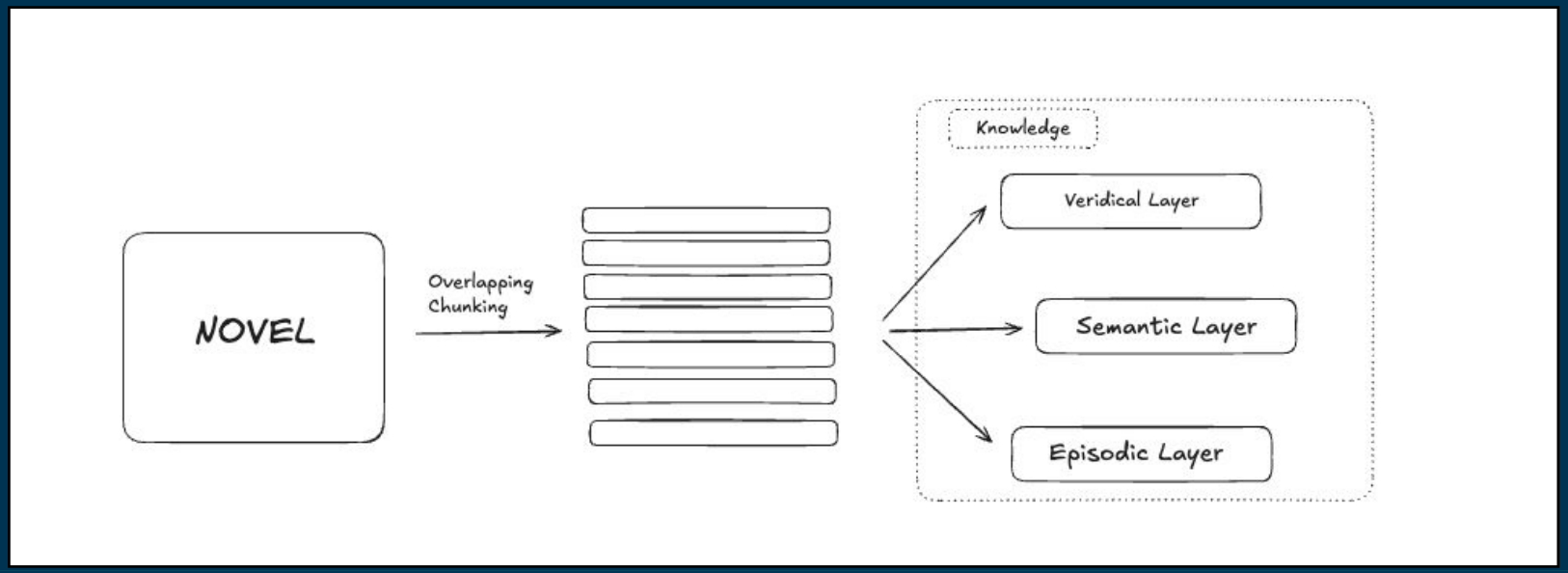
A backstory may agree with some passages but contradict evidence spread across distant chapters.

Example: Retrieved scenes support the backstory, but later chapters explicitly negate it.

Our Approach:

The Hierarchical Knowledge Source

- Veredical: Veridical Layer: Grounding in Factual Evidence
- Semantic: Abstracting Thematic Structure.
- Episodic: Reconstructing Narrative Flow

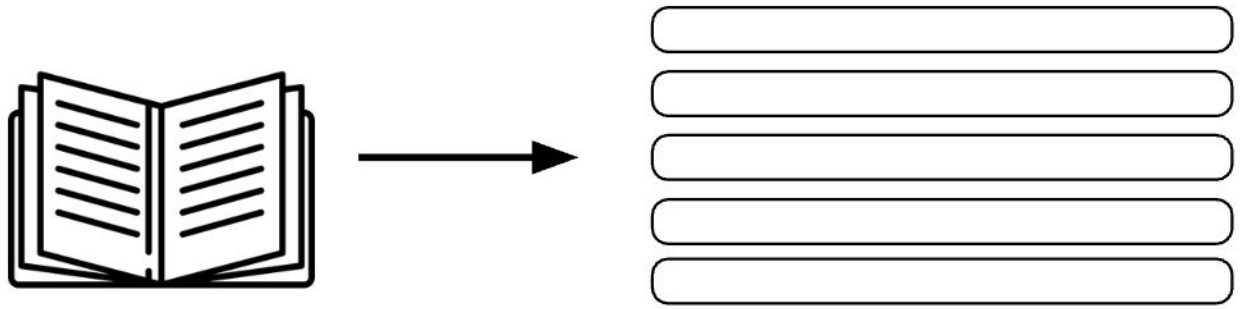


1. Veredical Layer

- Stores explicit, ground-truth facts from the novel
-
- Represented as subject–predicate–object triplets
- Organized as a global knowledge graph (HippoRAG style)
- Enables multi-hop factual reasoning across the story
- Retrieval is structure-driven, not similarity-driven
- Detects hard factual contradictions

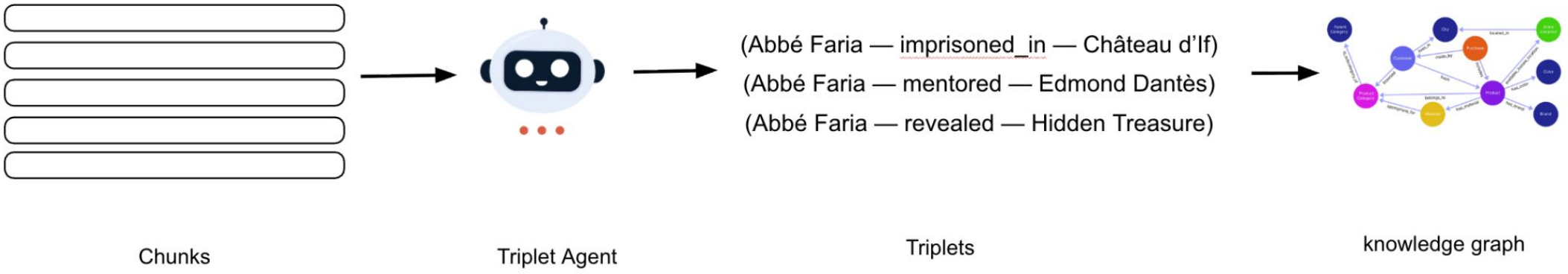
Veredical Layer Implementation

1. Overlapping Chunking



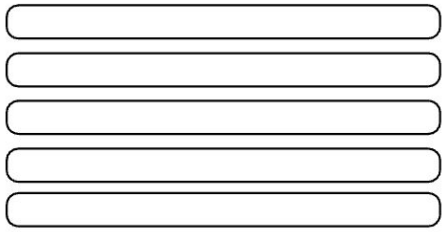
Veredical Layer Implementation

2. Extracting all the triplets and creating a knowledge graph



Veredical Layer Implementation

3. Creating Vector Embeddings for Entities and chuncks and store in Pathways vector db



[Harry, Snape, Dubeldore.....]

[0.1,2.0,4.0,5.0,6.0.7.0]
[0.1,2.0,4.0,5.0,6.0.7.0]
[0.1,2.0,4.0,5.0,6.0.7.0]
[0.1,2.0,4.0,5.0,6.0.7.0]....

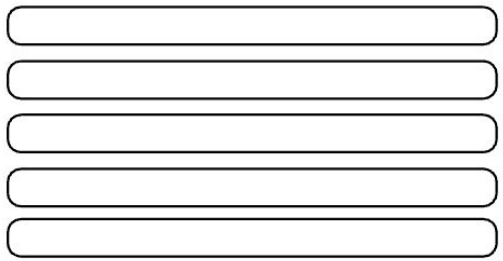
2. Semantic Layer

- Captures high-level themes and concepts across long-range context
- Handles meaning beyond individual text chunks
- Inspired by RAPTOR's hierarchical semantic abstraction
- Uses embeddings + GMM-based clustering
- Recursively summarizes similar chunks into a hierarchy
- Forms a semantic summary tree (local → global themes)
- Stores abstract narrative concepts and patterns
- Enables conceptual-level retrieval
- Helps detect semantic mismatches in backstory consistency

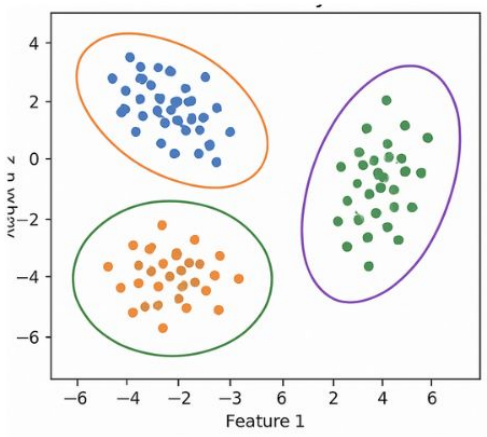
Novel Chunks

Semantic Layer Implementation

1. Gmm clustering on the basis of embedding semantic similarity



[0.1,2.0,4.0,5.0,6.0.7.0]
[0.1,2.0,4.0,5.0,6.0.7.0]
[0.1,2.0,4.0,5.0,6.0.7.0]
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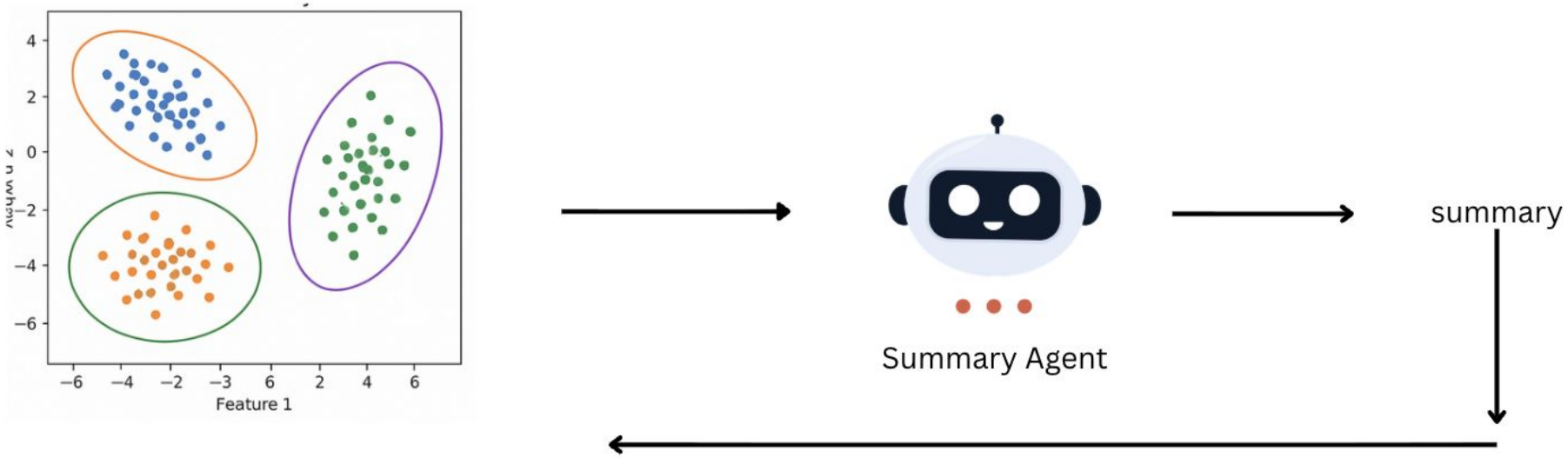


Chunks

GMM clusters

Semantic Layer Implementation

- 2. Summary for each cluster is generated together, and in tree structure

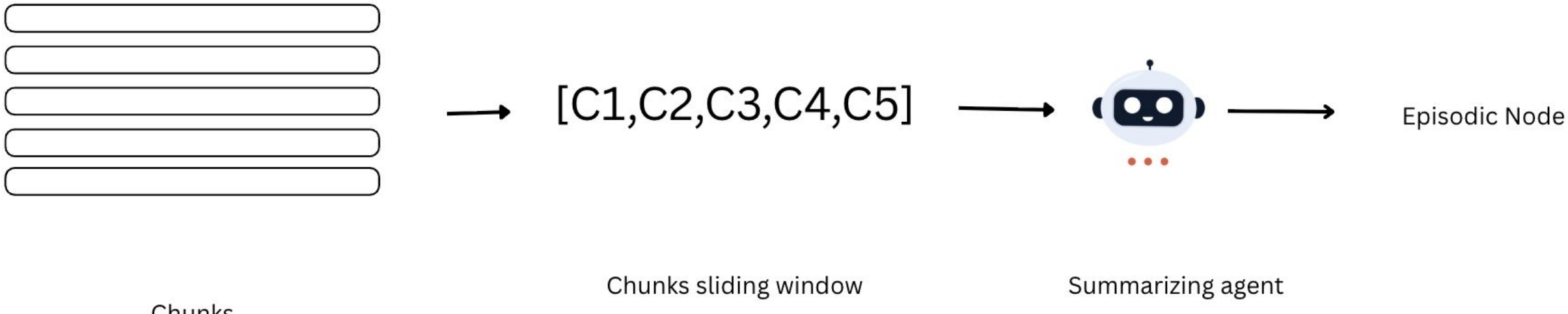


3. Episodic Layer

- Captures sequential narrative flow and plot progression
- Models temporal development missing from factual layers
- Preserves long-range causal chains across the story
- Built via sliding-window summarization over text chunks
- Each node represents a chronologically ordered episode
- Window size adapts to document length (stepped for short texts, logarithmic for long texts)
- Summaries retain key events and cause–effect relations
- Nodes are ordered to reconstruct the overall story arc
- Can be recursively applied to form higher-level narrative views

Novel Chunks

Episodic Layer Implementation



Storage and Retrieval

All memory layers stored in **Pathways Vector Database**
Supports:

- High-dimensional embedding search
- Fast iterative retrieval

Enables:

- Persistent memory across reasoning cycles
- Efficient access during probing

Iterative Probing Loop

- If memory is insufficient:
  a. Agent generates probing queries
  b. Queries split into focused sub-questions

- System retrieves:
  a. New facts
  b. Additional clusters
  c. Missing timeline events

- Memory is updated
- Reasoning is re-evaluated
- Loop continues until confidence is reached

Brain-Inspired Reasoning (PFC Analogy)

Inspired by **Prefrontal Cortex (PFC)**

Central agent:

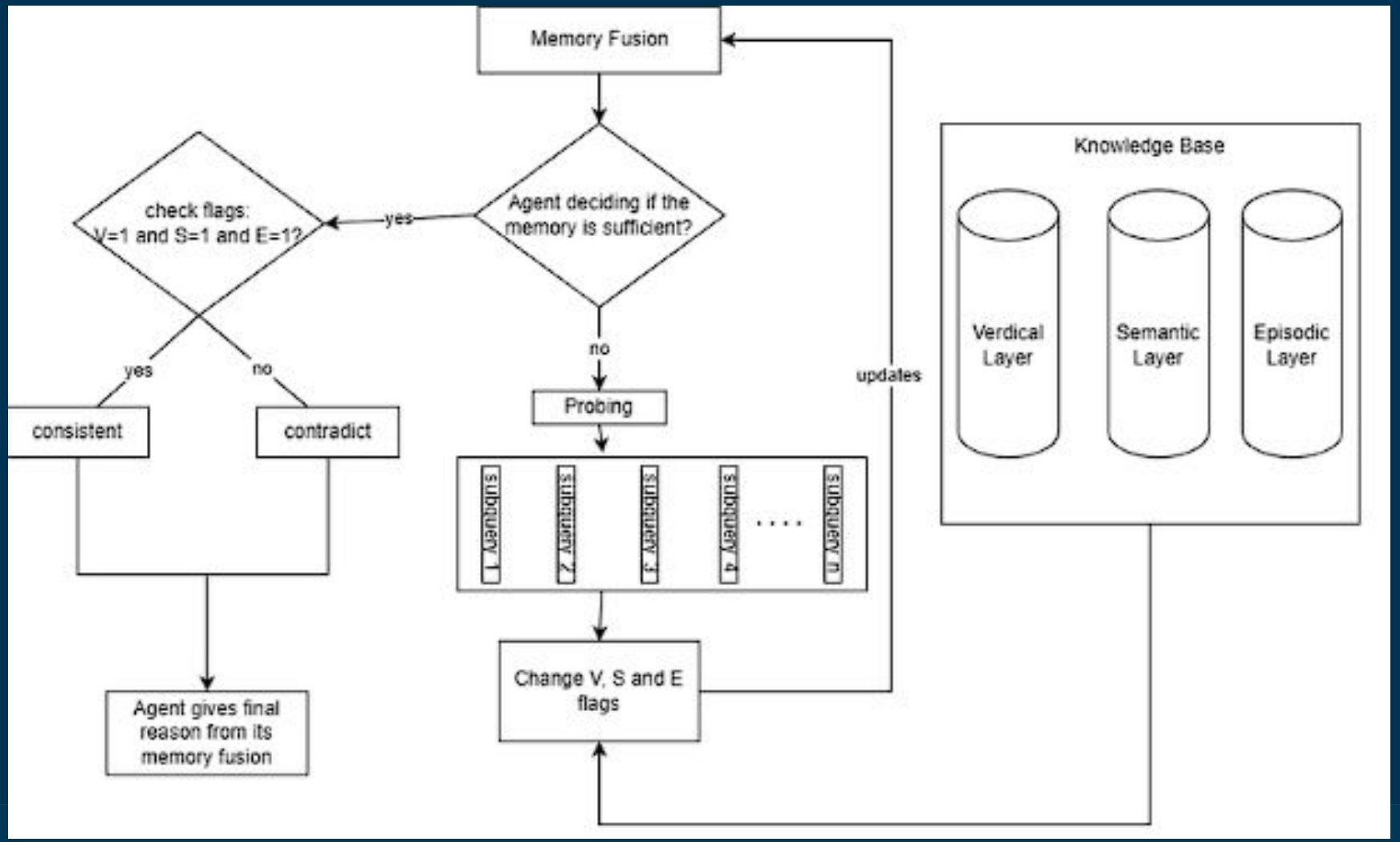
- Evaluates memory sufficiency
- Coordinates retrieval and reasoning

Final decision rule:

$$\text{Consistency} = f(V, S, E)$$

- V → factual consistency
- S → semantic alignment
- E → temporal validity

Retrieval Architecture



Why Our Approach Beats Standard RAG and LLM Approach?

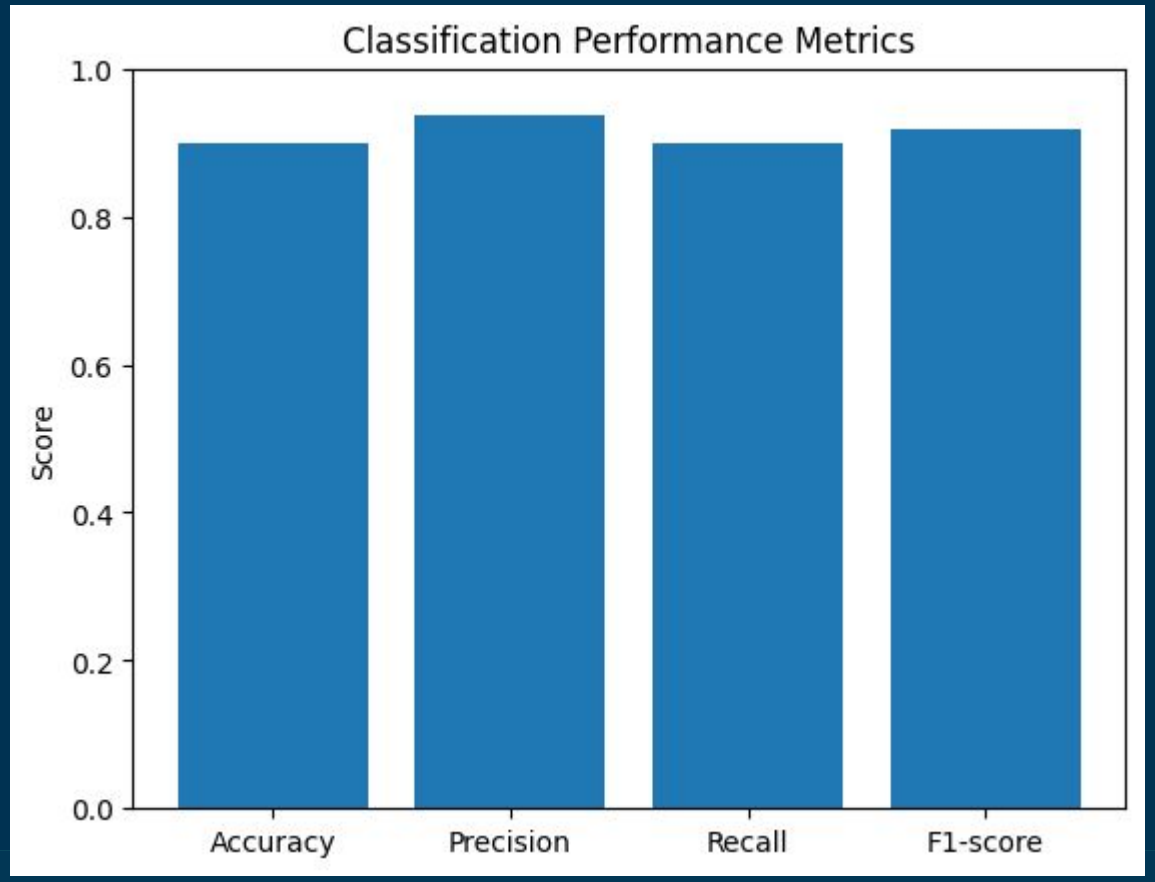
Standard RAG:

- One-shot
- Stateless
- Local reasoning

Our system:

- Iterative
- Memory-driven
- Global narrative reasoning
- Mimics human rereading and reflection

RESULTS AND OUTCOMES



Metric	Value
Accuracy	90.0%
Precision	93.75%
Recall	90.0%
F1-Score	91.8%

Limitations

- Embedding similarity may miss subtle causal cues
- Chunking can fragment long dependencies
- Implicit motivations remain hard to infer
- Iteration budget limits exhaustive reasoning

Conclusions

Narrative consistency requires:

- Memory
- Iteration
- Causal reasoning

Our framework:

- Models human-like reading
- Handles long context effectively
- Outperforms static RAG and direct LLM prompting

Overall: Demonstrates a scalable approach to global narrative reasoning

THANK YOU

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